

# MATHEUS STOLET

mstolet@mpi-sws.org

stolet.github.io

## EDUCATION

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### PhD, Computer Science

*Oct. 2021 - Oct. 2026*

Max Planck Institute for Software Systems

Advisor: Dr. Antoine Kaufmann

### MSc, Computer Science

*Sep. 2019 - Aug. 2021*

University of British Columbia

Thesis: Large Scale Federated Analytics and Differential Privacy Budget Preservation

Advisors: Dr. Ivan Beschastnikh and Dr. Aline Talhouk

### BA, Computer Science - Minor, Philosophy

*Sep. 2015 - May 2019*

University of British Columbia

## EMPLOYMENT

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### Research Intern, Microsoft Research Redmond

*Jun. 2024 - Sep. 2024*

- Implemented a library to make kernel-bypass networking more practical in cloud environments

- Optimized the Demikernel libOS for better performance

### Research Assistant, BC Cancer Research Centre

*May 2019 - Aug. 2019*

- Built a distributed analytics platform for executing queries across hospitals and research centres

- Applied differential privacy techniques to reduce information leakage from distributed queries

### Software Engineering Intern, Thrive Health

*May 2018 - Aug. 2018*

- Built a software module to support pre-surgical patient triage

- Developed frontend features for a web application using React and Redux

- Built backend infrastructure using AWS Lambda and SQS to scale media transcoding workflows

## TEACHING

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### Graduate Teaching Assistant, Max Planck Institute for Software Systems

- Cloud Reliability

*Apr. 2025 - Jul. 2025*

### Graduate Teaching Assistant, Max Planck Institute for Software Systems

- Operating Systems

*Apr. 2024 - Sep. 2024*

### Graduate Teaching Assistant, University of British Columbia

- CPSC 416: Distributed Systems

*Jan. 2020 - May 2020*

### Undergraduate Teaching Assistant, University of British Columbia

- CPSC 317: Internet Computing

*Sep. 2018 - Dec. 2018*

## PAPERS

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**Matheus Stolet**, Simon Peter, Antoine Kaufmann. Rethinking Polling Efficiency in Service Core Network Stacks. *Under Review* (pre-print arXiv:2607.16408).

**Matheus Stolet**, Simon Peter, Antoine Kaufmann. Chamelio: A Fast Shared Cloud Network Stack

for Isolated Tenant-Defined Protocols. *Under Review* (pre-print arXiv:2604.22603).

**Matheus Stolet**, Liam Arzola, Simon Peter, Antoine Kaufmann. Tail Contagion: Sub-microsecond Time Protection in Shared Software Network Datapaths. *Under Review* (pre-print arXiv:2309.14016).

Vaastav Anand, **Matheus Stolet**, Jonathan Mace, Antoine Kaufmann. Generating Representative Macrobenchmark Microservice Systems from Distributed Traces with Palette. In *APSys* 2025

Vaastav Anand, Zhiqiang Xie, **Matheus Stolet**, Roberta De Viti, Thomas Davidson, Reyhaneh Karimipour, Safya Alzayat, Jonathan Mace. The Odd One Out: Energy is not like Other Metrics. In *HotCarbon* 2022

Vaastav Anand, **Matheus Stolet**, Thomas Davidson, Ivan Beschastnikh, Tamara Munzner, and Jonathan Mace. Aggregate-driven trace visualizations for performance debugging. *arXiv:2010.13681* 2020

## SKILLS

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### Programming Languages

C, Python, Go, and JavaScript

### Technical Skills

DPDK, eBPF, kernel bypass, and differential privacy

### Languages

English (fluent) and Portuguese (native)

## AWARDS

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|-------------------------------------------------------------------------------------------|-------------|
| <b>1st Place ACM Student Research Competition</b> , SOSP                                  | <i>2023</i> |
| <b>International Student Tuition Award</b> , University of British Columbia               | <i>2019</i> |
| <b>Dean's List</b> , University of British Columbia                                       | <i>2016</i> |
| <b>Trek Excellence Scholarship</b> , University of British Columbia                       | <i>2016</i> |
| <b>Faculty of Arts International Student Scholarship</b> , University of British Columbia | <i>2016</i> |

## SERVICE

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|--------------------------------------------------------------------------|-------------|
| <b>Reviewer</b> , MPI Pre-Submission Application Review (PAR) Program    | <i>2023</i> |
| <b>Volunteer</b> , SOSP                                                  | <i>2023</i> |
| <b>Organizer</b> , Cornell, Maryland, Max Planck Research School (CMMRS) | <i>2022</i> |